PEER REVIEW HISTORY

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ARTICLE DETAILS

TITLE (PROVISIONAL)	How Defensive Medicine is Defined in European Medical Literature: A Systematic Review
AUTHORS	Baungaard, Nathalie; Skovvang, Pia Ladeby; Assing Hvidt, Elisabeth; Gerbild, Helle; Kirstine Andersen, Merethe; Lykkegaard, Jesper

VERSION 1 – REVIEW

REVIEWER	Mira, José Miguel Hernandez University of Elche, Health Psychology
REVIEW RETURNED	11-Sep-2021

GENERAL COMMENTS	This is a systematic review of defensive medicine that has been carried out according to the recommendations for this type of study. The study protocol has already been published. The content is in line with PRISMA although the subsection on Synthesis methods is not included in the text (the study protocol lists it on pages 8-9). I think this point should be described in more detail. It is a work with a clear objective, executed as intended. It provides information to delimit the concept of DM.

REVIEWER	Papanikitas, Andrew University of Oxford
REVIEW RETURNED	06-Oct-2021

GENERAL COMMENTS	I recommend acceptance with a couple of tiny corrections. Page 7 line 18 -despite incomplete (does not cover all medical practice) state-backed indemnity, the UK medical litigation system is generally NOT no-fault as in fault or negligence still needs to be determined in order for compensation to be awarded to someone who is harmed as a result of healthcare. And on p.8 line 42 it may be would being explicit that UK is included despite its BREXIT status as part of the European continent. In the discussion it might be worth reflecting whether the UK health system is closer to Canada or New Zealand or to other European countries in terms of funding and incentive structure. Finally a general reflection is to ask whether any specialty emerged as a dominant source of research data about defensive medicine in empirical studies, though looking at the table this may be hard to determine.

VERSION 1 – AUTHOR RESPONSE

Reviewer 1:
Dr. José Mira, Miguel Hernandez University of Elche, Salud Alicante-Sant Joan Health District.
1. This is a systematic review of defensive medicine that has been carried out according to the recommendations for this type of study. The study protocol has already been published. The content is in line with PRISMA although the subsection on Synthesis methods is not included in the text (the study protocol lists it on pages 8-9). I think this point should be described in more detail.
It is a work with a clear objective, executed as intended. It provides information to delimit the concept of DM.
Response:
We thank the reviewer for the positive and constructive comments.
The following changes have been made to the manuscript:
Methods p. 8, line 24 – p. 9, lines 1-14.
Data synthesis
For each study, the stated definition of DM was reviewed and assessed by all the six researchers. The stated definitions were extracted if they comprised constructions such as: “DM is…”, “DM is defined as…”, “DM refers to…”, or “DM is characterized by…”. If a study did not explicitly state a definition of DM, an interpretation of the study’s introduction to DM was made and excerpts to support the interpretation were extracted. If a study’s definition of DM was stated with references, these references were recorded and, by chain searching, followed back to the original source. The stated definitions of DM were categorized according to the included actions (e.g., “deviation from sound medical practice”) and motivations (e.g., “fear of lawsuit”) using qualitative content analysis(30). Next, any motives regarded as defensive were identified in the text, tables, figures as well as in the data collection methods in order to examine whether they differed from the motives stated in the study's definition of DM. Studies where researchers differed in the extraction and categorisation of DM motives were discussed among all researchers sometimes leading to rephrasing, merger, or de novo creation of categories. This was an iterative process until consensus could be reached.

Reviewer 2:
Dr. Andrew Papanikitas, University of Oxford.
1. I recommend acceptance with a couple of tiny corrections.
Response:
Thank the reviewer for the comments and for recommending the review for publication.

2. Page 7 line 18 - despite incomplete (does not cover all medical practice) state-backed indemnity, the UK medical litigation system is generally NOT no-fault as in fault or negligence still needs to be determined in order for compensation to be awarded to someone who is harmed as a result of healthcare.
The following changes have been made to the manuscript:
Introduction p. 5, lines 10-17.
However, contrary to the US, malpractice litigation is rare in many European countries, such as the Netherlands(15, 16), Denmark(7), Switzerland(17) and the UK(18). The medicolegal systems in these European countries do not hold physicians financially liable for malpractice or other treatment-related adverse events. Furthermore, in some European countries patients entitled to it are compensated for avoidable injuries by the government not requiring prove of health care provider negligence(19-21). This is known as a no-fault system. Nevertheless, DM is frequently reported in Europe and a substantial part of research on DM originates from Europe(6, 7, 15, 18, 22-24).

3. And on p.8 line 42 it may be would being explicit that UK is included despite its BREXIT status as part of the European continent.

Response:
Prior to our study, we discussed whether our systematic review should include data from Europe or European Union. Our systematic review includes data from European countries whether they are a part of the European Union or not. We deliberately chose this solution in order to include data from the UK despite its BREXIT status. We have rephrased inclusion criteria six to clarify this.
The following changes have been made to the manuscript:
Methods, Eligibility criteria p. 7, line 14.
6. The study includes data from Europe.

4. In the discussion it might be worth reflecting whether the UK health system is closer to Canada or New Zealand or to other European countries in terms of funding and incentive structure.
The following changes have been made to the manuscript:
Discussion, Future research p. 18, lines 16-23:
The geographic delimitation to Europe excluded countries like New Zealand and Canada that has medicolegal systems like that in the UK(73-75). DM studies from these countries are likely to deviate from the original, narrow definition of DM in ways similar to what we have demonstrated in the European studies. However, it is beyond the scope of this systematic review to identify and analyse the underlying medicolegal systems of countries worldwide. Investigating the interrelationship between medicolegal system and DM in future research could contribute to an understanding of how medicolegal systems influence the motives for practising DM.

5. Finally a general reflection is to ask whether any specialty emerged as a dominant source of research data about defensive medicine in empirical studies, though looking at the table this may be hard to determine.
The following changes have been made to the manuscript:
Results, Study selection and characteristics p. 10, lines 15-17.
The studies encompass 39 medical specialities with general practice (n = 14), obstetrics and gynaecology (n = 12), emergency department (n = 9), general surgery (n = 8) and anaesthesiology (n = 8) emerging as dominant sources of research data.

VERSION 2 – REVIEW

REVIEWER	Mira, José Miguel Hernandez University of Elche, Health Psychology
REVIEW RETURNED	10-Nov-2021

GENERAL COMMENTS	Authors have answered in a right manner. Changes improves the manuscripts. All are following the comments and questions of the referees.